

Feeding the City:
How Agricultural Productivity in Villages
Shapes Urban Structural Change

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Motivation & research question

- ▶ Government & development programs often target rural areas while cities offer higher standards of living ($\approx 4\%$ of public spending on agriculture)
- ▶ Classical theories focus on higher ag productivity releasing excess labor to cities, driving industrialization (Harris and Todaro, 1970; Lewis, 1954)
- ▶ Or do productivity gains retain workers in agriculture, lowering incentives to leave? $\implies$ ambiguous effect (Bustos et al., 2016; Glaeser, 2014; Matsuyama, 1992)
 - ▶ Scarce causal evidence on mechanisms: $\uparrow$ ag productivity $\implies ? \implies \uparrow$ urban growth (Bustos et al., 2020; Gollin et al., 2002; Lagakos et al., 2020)
- ▶ **Research Question:**

How does rising agricultural productivity in rural areas affect growth and structural transformation in cities in developing countries?

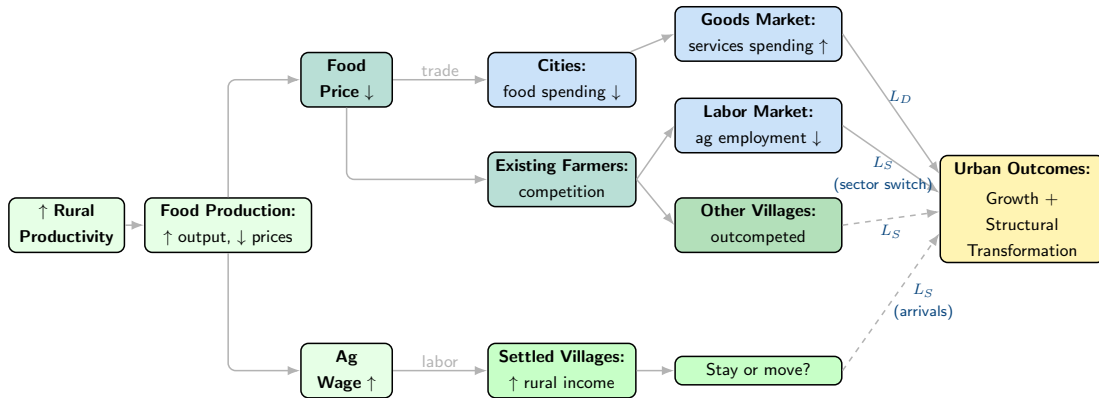
- ▶ **Setting:** Indonesia's Transmigration Program, a plausibly exogenous rural productivity shock
- ▶ **Main result:** modern employment +18% per SD of exposure; small insignificant pop effect
- ▶ **Not the settlers:** new arrivals do not account for the sectoral shift
- ▶ **Channels:**
 - Stronger effects in cities less connected to global markets (data)
 - Trade alone can reproduce the shift out of agriculture (while migration alone cannot)

Rural productivity transforms cities through **trade, not **migration****

1. **Rural-to-urban spillovers:** village shocks affect employment outcomes in cities
 - prior work mostly looked at within-region differences ([Bustos et al., 2016](#))
 - adds supply-side spillovers from local shocks through demand + capital ([Egger et al., 2022](#); [Moretti, 2010](#))
2. **Channels of trade vs. migration** ([Bryan and Morten, 2019](#); [Tombe and Zhu, 2019](#))
 - larger responses in cities less connected to global markets as in closed economies ([Matsuyama, 1992](#))
 - evidence through novel channels of spillovers (capital flows in Brazil in [Bustos et al. \(2020\)](#))

Trade-mig lit

Mechanism map: from rural productivity to city outcomes



The source of the rural productivity shock

↑ Rural
Productivity

First stage

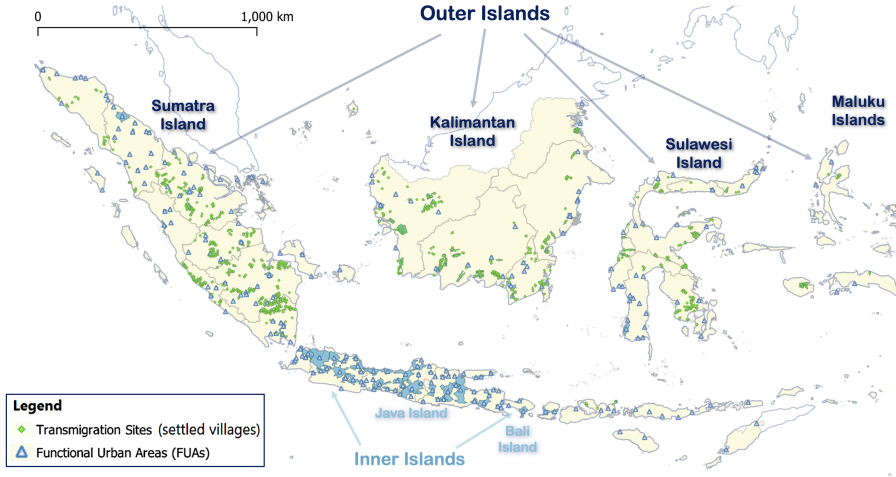
Setting: a rural productivity shock and observed urban outcomes

- ▶ Indonesia's Transmigration (TM) Program resettled 2+ million people in 1979-1988
 - ▶ Free 2-ha land lots with a house + free transportation for families to encourage moving to Outer Islands
 - ▶ Aimed at reducing overpopulation on Java and Bali and improving food security
 - ▶ Migrant destinations were plausibly as good as random
- ▶ Agro-climatic similarity between migrant origins and destinations (soil, climate, geography) is quasi-random & a good proxy for migrant skills and productivity ([Bazzi et al., 2016](#))
- ▶ I construct urban exposure to rural productivity and measure urban outcomes



Sample Destination Village

Setting: transmigration villages scattered across the Outer Islands



[Back](#)

Large variation in agro-climatic conditions in receiving communities!

Setting: agro-climatic similarity as a source of rural productivity

Inner Islands

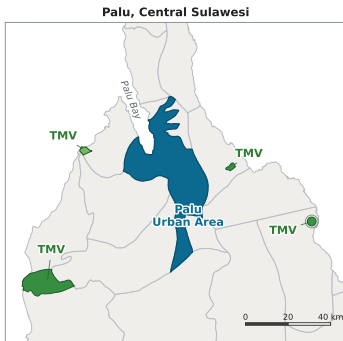


Outer Islands

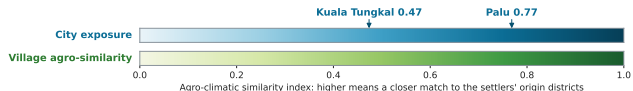
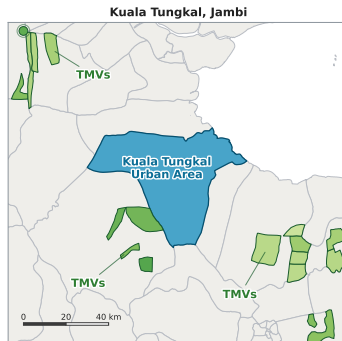


Sample regions: urban exposure to village productivity varies widely

A City with **High** Ag Productivity Shock



A City with **Low** Ag Productivity Shock



Productivity effect *conditional on comparable settler count + agro-climatic village characteristics*

AgSim definition

Exposure definition

Travel time

Spread

$$UrbanOutcome_c^{2010} = \beta AgSim_c + \gamma Settlers_c + \nu AvgVillageGeo_c + \kappa UrbanBaseline_c^{1980} + \epsilon_c$$

- ▶ β : the causal effect of a productivity shock in the surrounding villages conditional on controls
- ▶ **Identification assumption**: conditional on settler numbers, village agro-climate and the 1980 baselines, match quality is plausibly as good as random
- ▶ Exposure is uncorrelated with most 1980 baselines; the exceptions are controls

Data

Placebo

Borusyak-Hull

Balance: exposure is uncorrelated with most 1980 baselines

	Population + Employment		Agric Empl		Modern Empl		Baseline birthplace shares	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	ln pop	ln emp	ln L agric	agric %	ln L modern	modern %	% Born Outer Isl	% Born Java/Bali
TMV AgSim (z, 300min)	-0.051 (0.113)	-0.067 (0.100)	-0.164* (0.092)	-0.039 (0.032)	-0.066 (0.180)	0.022 (0.031)	0.020 (0.014)	-0.020 (0.014)
Transmigrant count (IDW)	✓	✓	✓	✓	✓	✓	✓	✓
Agro-Clim. TMV controls	✓	✓	✓	✓	✓	✓	✓	✓
Mean dep var	11.084	9.837	9.235	0.558	8.796	0.385	0.925	0.075
Baseline empl share			0.558		0.385			
Std dev dep var	0.99	0.96	1.06	0.22	1.31	0.20	0.11	0.11
Observations	134	134	134	134	134	134	134	134

7 of 8 baseline levels insignificant (unbalanced added as control)

Coefplot

By sector

Result 1: Cities change their sectoral employment without pop growth

	Population + Employment			Agric Empl		Modern Empl	
	(1) ln pop	(2) ln emp	(3) unemp rate	(4) ln L agric	(5) agric %	(6) ln L modern	(7) modern %
Exposure to village productivity (1 SD)	0.060 (0.059)	0.033 (0.056)	0.009 (0.007)	-0.167* (0.096)	-0.053*** (0.018)	0.169*** (0.061)	0.068*** (0.015)
Transmigrant count (IDW)	✓	✓	✓	✓	✓	✓	✓
Baseline Pop 1980	✓	✓	✓	✓	✓	✓	✓
Urban share 1980	✓	✓	✓	✓	✓	✓	✓
Total Empl 1980	✓	✓	✓	✓	✓	✓	✓
Agri Empl 1980	✓	✓	✓	✓	✓	✓	✓
Agro-Clim. TMV controls	✓	✓	✓	✓	✓	✓	✓
Mean dep var	11.839	10.930	0.123	9.816	0.374	10.229	0.517
Std dev dep var	0.858	0.843	0.041	0.898	0.184	0.979	0.157
Observations	133	133	133	133	133	133	133

⇒ New arrivals cannot account for the shift

How large

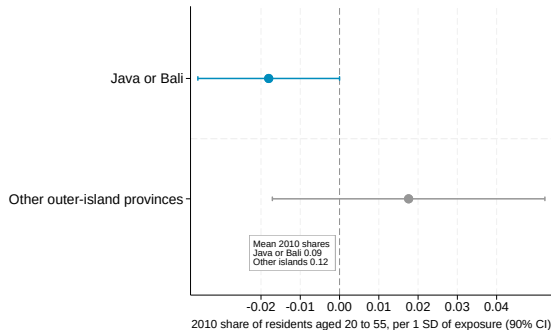
Cohorts

Decomposition

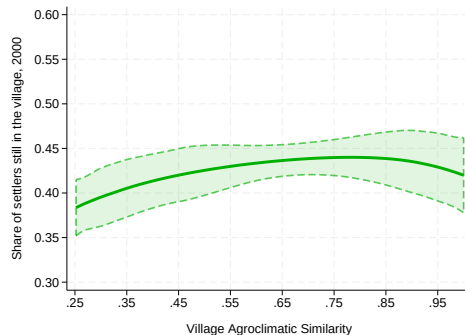
Spec curve

Result 2: Transmigrant settlers are not moving differentially

Cities: 2010 residents aged 20 to 55 by birthplace



Villages: share of settlers still there in 2000

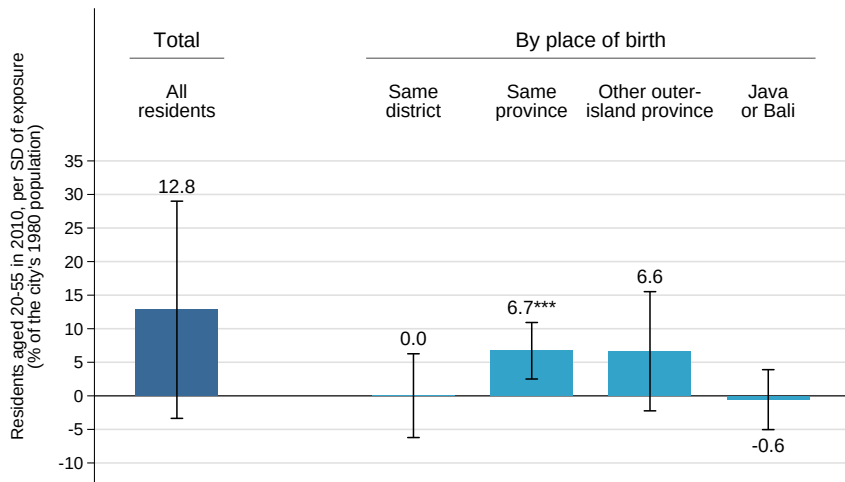


► Fewer city residents born in Java or Bali, the closest proxy for Transmigration settlers

Stay-share

Migration

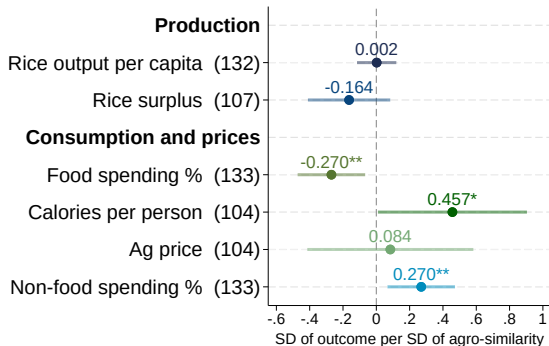
Result 2: population change by place of birth



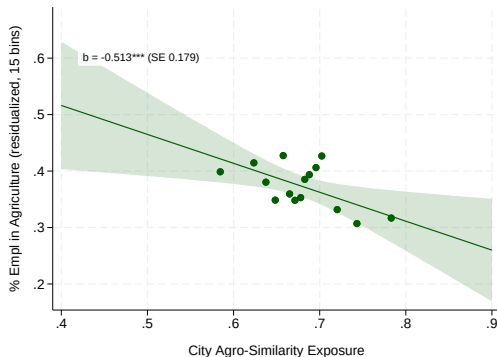
Main Spec, Conley 200 km, 90% CIs, N = 133 cities; residents born 1955-1990; the four birthplace bars add up to the total

Result 3: Food expenditures fall while consumption does not

Cities: goods and labor markets



Cities: agricultural share *falls*



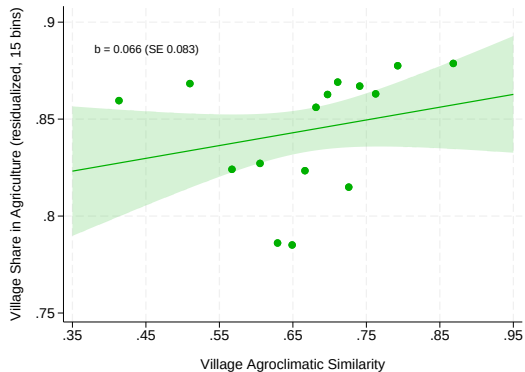
⇒ Spending shifts away from food, freeing up income for other sectors ⇒ $\uparrow L_D$ in non-ag

Budget

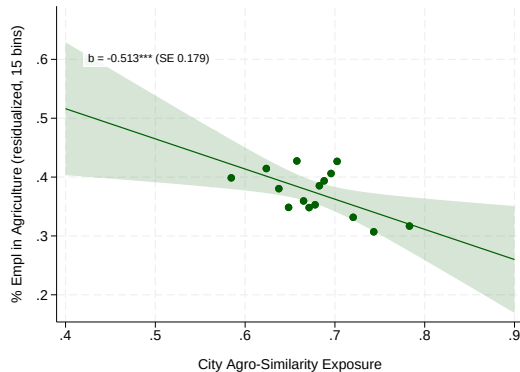
Prices

Result 3: the agricultural share falls in nearby cities and is flat to rising in villages

Villages: agricultural share *flat to rising*



Cities: agricultural share *falls*



The same gain leaves the village agricultural share flat to rising and lowers it in the nearby cities

Semipar view

Result 4: the response is larger in trade-closed cities

A city that can already import food cheaply gains little from a better hinterland ([Matsuyama, 1992](#))

	(1) ln pop	(2) ln emp	(3) unemp rate	(4) ln L agric	(5) agric %	(6) ln L modern	(7) modern %
AgSim, more closed cities: β_1	0.130 (0.086)	0.113 (0.082)	0.003 (0.009)	-0.152 (0.132)	-0.065*** (0.024)	0.283*** (0.090)	0.093*** (0.022)
AgSim, more open cities: $\beta_1 + \beta_2$	0.020 (0.057)	-0.011 (0.059)	0.012* (0.007)	-0.177* (0.096)	-0.046** (0.021)	0.107 (0.067)	0.054*** (0.016)
High openness (=1)	0.023 (0.067)	0.005 (0.082)	-0.005 (0.011)	0.044 (0.150)	0.003 (0.028)	-0.012 (0.084)	-0.013 (0.019)
Mean dep var	11.839	10.930	0.123	9.816	0.374	10.229	0.517
Std dev dep var	0.858	0.843	0.041	0.898	0.184	0.979	0.157
Observations	133	133	133	133	133	133	133

Note: High openness = below-median route distance to the Thai and Vietnamese rice hubs.

⇒ **Hinterland productivity matters less for cities with easier access to open markets**

Openness

A two-sector spatial equilibrium of cities and their villages

- ▶ **Locations:** Multiple cities and rural areas around them, each with land and an amenity
- ▶ **Two sectors:** **food**, traded at a cost rising in travel time, and a **modern sector** (manufacturing + services)
- ▶ Workers choose a **location** (elasticity θ_L) first, then a **sector** (elasticity θ_S)
- ▶ Migration is costly, with the cost estimated from IFLS household migration gravity

Stone-Geary Utility

$$U = (c_A - \bar{a})^{\psi_A} c_M^{1-\psi_A}$$

- ▶ $\bar{a}$ is a subsistence food floor (enables ag contraction in cities)
- ▶ Only income above it is split by shares
- ▶ Cheaper food, smaller food share

Model Goals

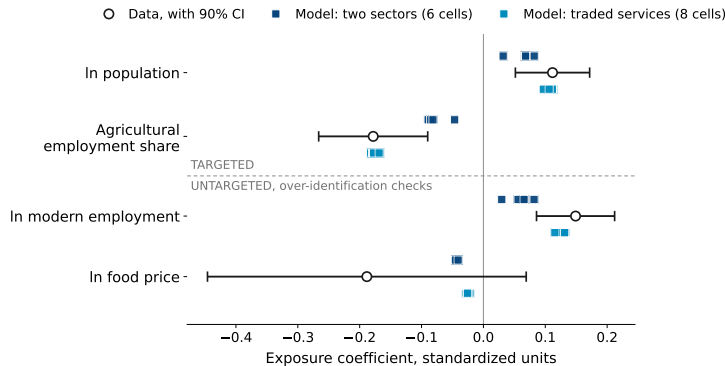
Equilibrium

Stone-Geary

Migration friction

Model fit: only traded services matches both targeted moments

Estimation: run the main regression on model and real data $\implies$ calibrate θ_L, θ_S for targeted moments



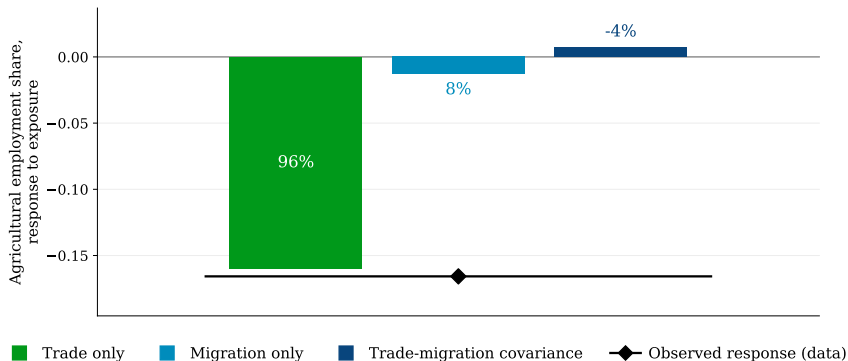
Data markers: the calibration-panel estimates the model was fit to (128 cities)

- Model fits the data well, with better predictions with tradable service component

Counterfactuals: the trade-only economy reproduces most of the contraction

- Counterfactuals: turn off migration/trade $\Rightarrow$ re-solve the model to get new eq'm

How



$\Rightarrow$ **Trade** suffices to explain agricultural contraction

All panels

Rural productivity transforms cities through trade, not migration

- ▶ **Rural-to-urban spillovers exist:** a policy aimed at villages reshaped the cities it never targeted
- ▶ **Modern sector expands** in levels and in shares, with no detectable change in population or unemployment
- ▶ **Food trade** enables structural change as the food budget share falls, freeing urban workers & spending, especially in cities less connected to global trade
- ▶ $\implies$ Ag productivity gains can translate into expanding services and manufacturing in cities without substantial migration

Thank you!



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The policy lever: food trade costs

- ▶ The policy lever is the **food trade cost** between a city and its hinterland, the roads-and-ports margin
- ▶ The mechanism needs a large urban agricultural margin and has its largest effect where links to outside food markets are weak
- ▶ Relevant for considering the secondary cities of Sub-Saharan Africa and South Asia, where the coming urban growth will land ([Henderson and Turner, 2020](#))

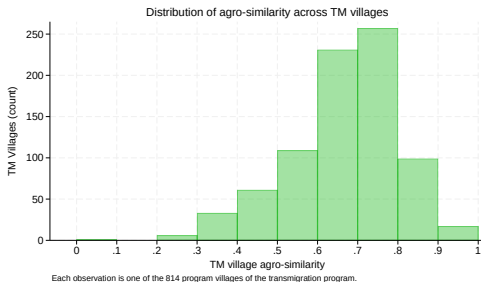
Where food is already cheap to import, the modern-sector response is smaller

Openness measures

◀ back

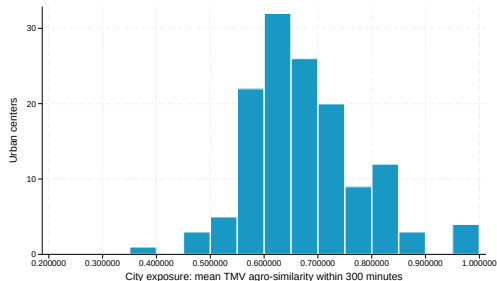
Variation in urban exposure to TM village agricultural productivity

Village-Level Agro-Similarity



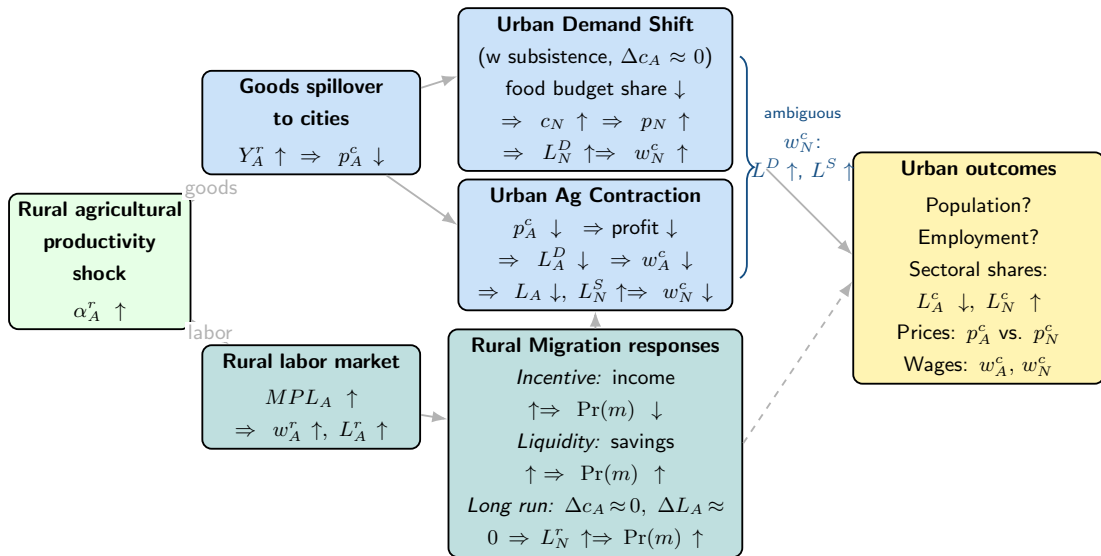
- ▶ Villages: 75% have at least some rice production
- ▶ 1 st. dev. $\uparrow$ similarity $\implies$ 20% $\uparrow$ rice productivity

Urban Center Exposure to TMV Agro-Similarity



- ▶ Rice consumption: $> 25\%$ of all HH expenses on agricultural goods in cities

Detailed mechanism map: from rural productivity to city outcomes



TM village agricultural productivity proxied by agro-climatic similarity among migrants

- ▶ Rural migrants' productivity in destinations depends on their skills gained in places of origin
- ▶ Agro-climatic similarity captures how similar climate, soil, geography, etc. are in origin districts vs. destination villages among migrants (AS) ([Bazzi et al., 2016](#)):

$$\text{Agroclimatic Similarity}_v = - \sum_{o=1}^{\text{Origins}} \pi_{ov} d(x_o, x_v) \quad (1)$$

- ▶ x_o, x_v = a vector of agro-climatic conditions (climate, soil, geography) in origin districts o and destination villages v
- ▶ $d(x_o, x_v)$ = agro-climatic distance d between district of origin o and destination village v
- ▶ π_{ov} = share of migrants residing in TM village v who were born in origin district o
- ▶ -1 to reverse the dissimilarity “distance” measure into a similarity measure

[Back to Sample Regional Map](#)

City exposure to TM village agricultural productivity (main X)

- Exposure averages agro-similarity (AS_v) equally over TM villages within 300 minutes:

$$\text{Ag Similarity Exposure}_c = \frac{1}{|V_c|} \sum_{v \in V_c} AS_v \in [0, 1] \quad (2)$$

- V_c = TM villages within 300 travel minutes of city c , $\tau_{cv} \leq 300$
- τ_{cv} = least-cost travel time in minutes between city c and village v
- The settler count enters separately, as an inverse-distance-weighted control

[Back to Sample Regional Map](#)

Sample & main outcomes

133 urban centers in the estimation sample (137 lie within 300 travel minutes of a Transmigration village; 814 villages in all)

Outcomes of Interest	Geography	What Is In the Data
Population, Employment, and Migration from the Population Census (1980, 2010):		
Sectoral employment	Cities + Villages	Individual employment in agriculture, manufacturing, services
Population	Cities + Villages	Population level/growth; total employment level/growth
Migration (by flow and origin)	Cities + Villages	Migration rates (by birth region and region 5 years ago)
Prices, Consumption, & Wages from Susenas HH Survey (2002, 2011):		
Prices (ag vs. non-ag)	Cities	HH Consumption module: Unit values by food item, crop/food price indices, housing price index (<i>proxy for non-tradables' price</i>)
Consumption expenditures	Cities	Consumption by food (quantity & value) and non-food item (value), expenditure shares (food and non-food)
Wages (ag vs. non-ag/services)	Cities	Median wage by worker employment sector (real wages calculated by dividing by the corresponding price index)
Agricultural Production from PODES Village Survey (2003):		
Agricultural production	Cities + Villages	Production by crop/item, yield per land area

Robustness: an expected-exposure control keeps about three quarters of the headline

	agric %	ln modern
Headline	-0.053*** (0.018)	0.169*** (0.061)
Controlling for expected exposure	-0.042** (0.020)	0.125** (0.061)
Recentered, score swap	-0.017 (0.011)	0.045 (0.037)
Recentered, settler swap	-0.011 (0.013)	-0.006 (0.058)

Conley 200 km standard errors in parentheses; N = 133

- Exposure is recentered against the settler assignments that could have happened
 - The settler-swap treatment correlates with the headline one at only **0.26**
 - Under the settler swap the average city moves **42 places** in the exposure ranking of 133 cities
- Subtracting the expectation instead leaves neither recentered form distinguishable from zero**

Magnitude: 5.3 percentage points is over a quarter of the 30-year fall

- ▶ The average city had 55.9% of its workers in agriculture in 1980 and 37.4% in 2010
- ▶ A 1 SD exposure gap is 5.3 percentage points of that share, over a quarter of the 18.5 percentage point fall
- ▶ The yardstick is the 1980 to 2010 fall in the agricultural share
- ▶ These are local comparisons among Outer Island cities with transmigration hinterlands

Back

Balance: 12 of the 14 rows are indistinguishable from zero



Both flagged 1980 baselines are controls in every headline regression

Back

By sector

Balance by sector: two 1980 employment levels are flagged at 10%

	Population + Employment		Agric Empl		Manuf Empl		Serv Empl		Baseline birthplace shares	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	ln pop	ln emp	ln L agric	agric %	ln L manuf	manuf %	ln L serv	serv %	% Born Outer Isl	% Born Java/Bali
TMV AgSim (z, 300min)	-0.051	-0.067	-0.164*	-0.039	-0.365*	-0.014	-0.011	0.036	0.020	-0.020
	(0.113)	(0.100)	(0.092)	(0.032)	(0.207)	(0.010)	(0.182)	(0.026)	(0.014)	(0.014)
Transmigrant count (IDW)	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Agro-Clim. TMV controls	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Mean dep var	11.084	9.837	9.235	0.558	6.859	0.064	8.614	0.321	0.925	0.075
Baseline empl share			0.558		0.067		0.321			
Std dev dep var	0.99	0.96	1.06	0.22	1.53	0.06	1.30	0.17	0.11	0.11
Observations	134	134	134	134	128	134	134	134	134	134

Log agricultural employment is a control; log manufacturing employment (N 128) is not

[Back](#)
[Coefplot](#)

Backup: the headline, all seven outcomes

	Population + Employment			Agric Empl		Modern Empl	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	ln pop	ln emp	unemp rate	ln L agric	agric %	ln L modern	modern %
Exposure to village productivity (1 SD)	0.060 (0.059)	0.033 (0.056)	0.009 (0.007)	-0.167* (0.096)	-0.053*** (0.018)	0.169*** (0.061)	0.068*** (0.015)
Transmigrant count (IDW)	✓	✓	✓	✓	✓	✓	✓
Baseline Pop 1980	✓	✓	✓	✓	✓	✓	✓
Urban share 1980	✓	✓	✓	✓	✓	✓	✓
Total Empl 1980	✓	✓	✓	✓	✓	✓	✓
Agri Empl 1980	✓	✓	✓	✓	✓	✓	✓
Agro-Clim. TMV controls	✓	✓	✓	✓	✓	✓	✓
Mean dep var	11.839	10.930	0.123	9.816	0.374	10.229	0.517
Std dev dep var	0.858	0.843	0.041	0.898	0.184	0.979	0.157
Observations	133	133	133	133	133	133	133

- No effect on the unemployment rate, so cities are not growing at the expense of people being jobless
- The agricultural employment level falls with the share, at the 10% level

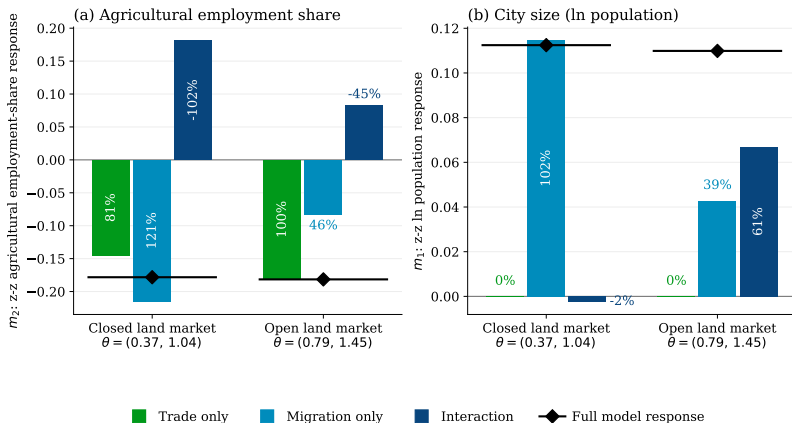
The three legs of the argument

1. **Rationalize.** Each city and its hinterland form a closed Matsuyama economy: villages specialize in food, the city in the modern sector
2. **Evidence on each channel.** Composition moves more than scale, villages keep farming, settlers stay, in-migrants come from the same province, plus the openness test
3. **Quantify, conditional on the model.** Shut trade, then migration, and report their interaction as its own term

The third leg is mechanism identification inside a model. It is never a claim about what fraction of the measured effect trade explains

Credit where it is due: the design descends from [Bustos et al. \(2016\)](#), and [Bustos et al. \(2020\)](#) follow agricultural windfalls through the banking system into industrial growth elsewhere

Both outcomes, both land-market closures



City size's trade-only response is zero by construction; the trade-only economy reproduces 81-100% of the agricultural-share response

Two further connected literatures

- ▶ **Agricultural technology's reach to cities:** gains followed into urban structure
 - ▷ ag-technology and extension studies measure farm yields and national growth ([BenYishay and Mobarak, 2019](#); [McArthur and McCord, 2017](#))
- ▶ **Settlement followed to nearby cities:** rural gains translate into structural change
 - ▷ the frontier-settlement literature follows the settlers ([Bazzi et al., 2016](#); [Pellegrina and Sotelo, 2025](#))

◀ back

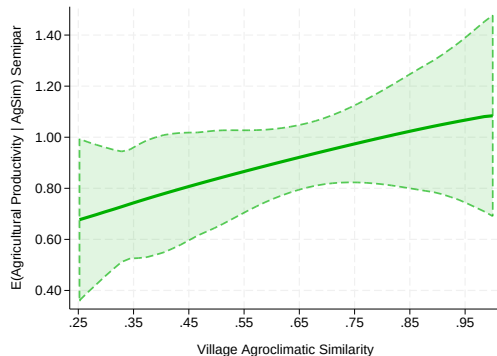
The source of each trade-vs-migration split

	What moves	Parameters pinned by	The split comes from
Tombe-Zhu 2019 China	● goods, internal + international ● workers, hukou costs	● costs backed out of trade flows, migrant stocks, wage gaps	● simulated cost cuts in the calibrated GE ● no shock identifies a response
Bryan-Morten 2019 Indonesia	● workers only ● one freely traded good	● migration costs estimated from migrant stocks + wages	● no trade margin, by construction ● all gains through moving people
Blanchard et al. 2025 African cities	● people travel to town ● services become tradable	● mobility costs from measured travel around cities	● counterfactual travel costs in the calibrated model
This paper Indonesia	● food + workers ● a friction on each	● two elasticities fit to reproduce the causal exposure coefficients ● untargeted moments as checks	● each channel switched off against the estimated response

Here the parameters are fit to the causal exposure coefficients; the shares describe the model's response

Backup: agro-similarity predicts village agricultural productivity

Village agricultural productivity against agro-climatic similarity



Same relationship, linear

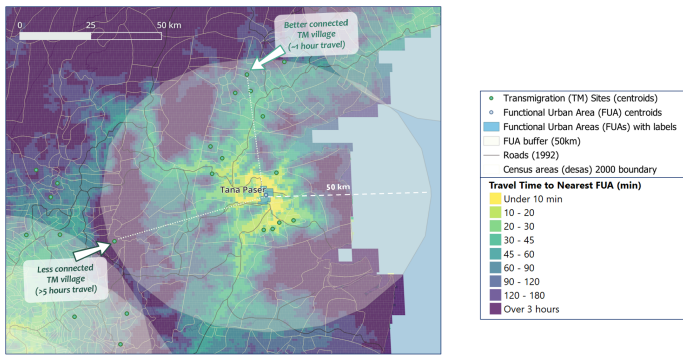
Village ag. productivity	
Agro-climatic similarity	0.525*** (0.201)
Villages	768

Conley 200km SE. *** 1%, ** 5%, * 10%.

Across the plotted range, village productivity rises by about 0.4 tons per hectare, against a median near 0.43

Backup: the land and sea travel-time surface

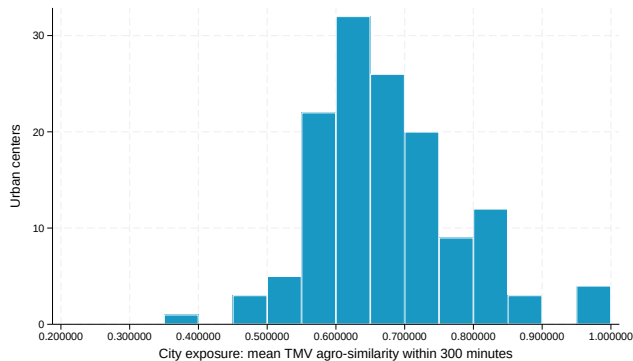
Villages equidistant **as the crow flies** are not equidistant **in travel time**



- Cost-distance is measured in **minutes**, over a friction surface combining terrain, land cover, the road network, and sea legs between islands
- The 300-minute window is a meaningful, achievable travel distance and keeps the closer three quarters of centers

Backup: exposure varies across the cities within 300 minutes

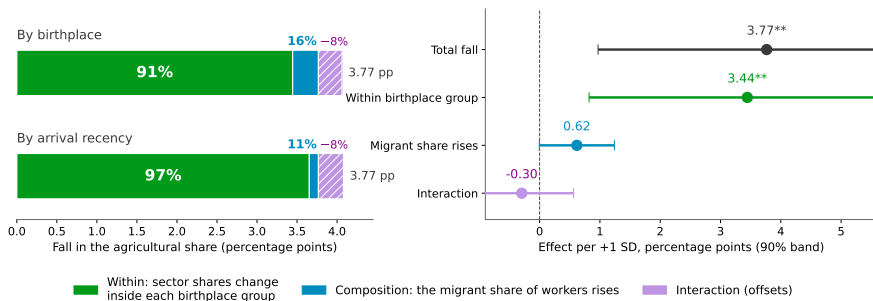
Distribution of city exposure across the urban centers within 300 minutes



The identifying comparison is between cities whose nearby villages happened to receive better- or worse-matched settlers, holding the number of settlers and the villages' own agro-climatic conditions fixed

Backup: the components of the fall in the agricultural share

Within margin, composition margin, and the exact remainder

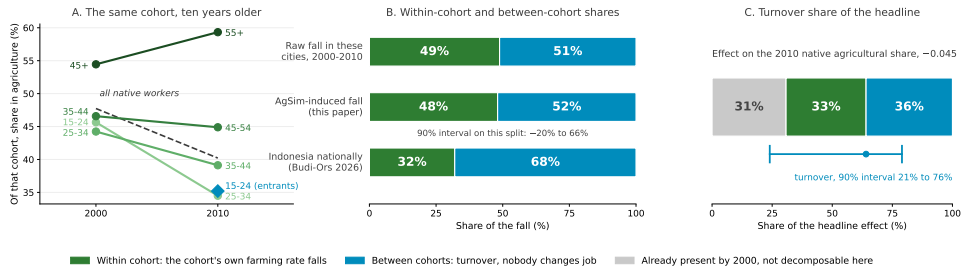


- ▶ The within component is 91% of the fall and composition 16%, offset by a small interaction
- ▶ Above the 80% that [Eckert and Peters \(2025\)](#) attribute to within-region change in the 1880-1920 United States
- ▶ The within component still mixes workers changing sector with one generation replacing another

Total measured on the 2010 birthplace-by-sector cross: 3.77 percentage points, against 5.3 in the headline table

Backup: turnover is established, within-cohort switching is undetermined

Native-born cohorts followed one age band forward, 2000 to 2010



The effect splits into near-equal thirds: in place by 2000, cohorts' own farming rate falling, and cohort turnover; the middle third is imprecise, so the censuses do not identify switching

Backup: the modern sector split into manufacturing and services

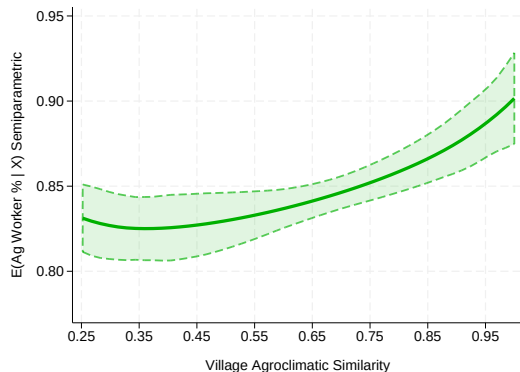
	Employment (Ln)		Employment shares	
	(1) ln L manuf	(2) ln L serv	(3) manuf empl %	(4) serv empl %
TMV AgSim (z, 300min)	0.250* (0.133)	0.157** (0.066)	0.010** (0.005)	0.058*** (0.018)
Transmigrant count (IDW)	✓	✓	✓	✓
Baseline Pop 1980	✓	✓	✓	✓
Urban share 1980	✓	✓	✓	✓
Total Empl 1980	✓	✓	✓	✓
Agri Empl 1980	✓	✓	✓	✓
Agro-Clim. TMV controls	✓	✓	✓	✓
Mean dep var	7.542	10.134	0.046	0.471
Baseline empl share	0.064	0.320	0.064	0.320
Std dev dep var	1.357	0.970	0.042	0.146
Observations	133	133	133	133

Conley 200 km standard errors, except column 1, heteroskedasticity-robust

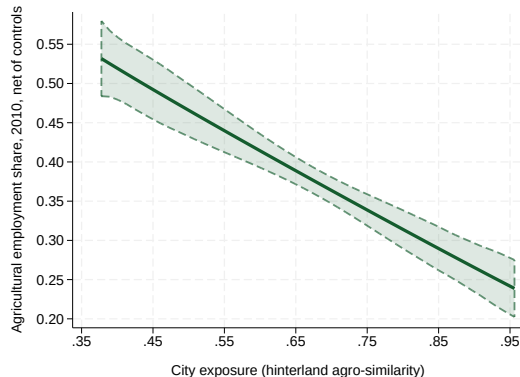
The analysis works with the modern sector as a whole throughout, following the Matsuyama framing in which the relevant margin is agriculture against everything else. The split is reported once, here, so the reader can see which component moves

Backup: both sides of the trade relationship, side by side

Villages: agricultural share flat to mildly rising



Cities: agricultural share falls



The agricultural share is flat to mildly rising across the villages and falls across the nearby cities: Matsuyama's two trade regimes, visible in the same setting

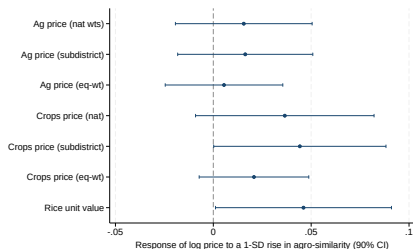
Backup: the budget and wage margins in their own units

	Prices & Spending		Wages		
	(1) nonfood %	(2) price housing	(3) ln wage nonag	(4) ln wage	(5) ln real wage
TMV AgSim (z, 300min)	0.013** (0.006)	-0.004 (0.111)	0.039 (0.032)	0.027 (0.034)	0.002 (0.031)
Transmigrant count (IDW)	✓	✓	✓	✓	✓
Baseline Pop 1980	✓	✓	✓	✓	✓
Urban share 1980	✓	✓	✓	✓	✓
Total Empl 1980	✓	✓	✓	✓	✓
Agri Empl 1980	✓	✓	✓	✓	✓
Agro-Clim. TMV controls	✓	✓	✓	✓	✓
Mean dep var	0.447	1.593	14.065	13.971	13.743
Std dev dep var	0.050	0.910	0.272	0.313	0.239
Observations	133	133	133	133	132

- ▶ Non-food expenditure share up 1.3 percentage points, at the 5% level; housing rents do not move detectably
- ▶ None of the three wage measures moves detectably
- ▶ The model must produce flat wages through offsetting price and land terms, with agglomeration at zero

Backup: why the measured food price is not the evidence

Price response across seven measures: small and positive, two of seven at the 10% level



- ▶ The delivered-price decline is model-implied, never measured, and two things stand in the way
- ▶ The price measures are household **unit values**, which embed quality upgrading: exposed cities buy less rice per capita while paying more per kilogram
- ▶ The 2002 to 2011 window brackets the national rice-price inflation that followed the 2004 import restrictions
- ▶ So the falling **budget share** is the robust demand-side object, and the measured price sign is not evidence against a locally cheaper delivered price

Backup: the stay-share holds under every definition

Numerator definition	Gradient on agro- similarity	Island- clustered std. error	Wild bootstrap p-value	Conley 200 km p-value	Corr. with the plotted measure
All Java/Bali-born residents (plotted measure)	0.078	(0.106)	0.69	0.57	1.000
Excluding arrivals in the previous five years	0.080	(0.093)	0.67	0.52	0.989
Excluding those born after settlement	0.071	(0.091)	0.67	0.54	0.995
Excluding both groups	0.080	(0.086)	0.66	0.49	0.985

The featured definition divides a village's 2000-census residents born in Java or Bali, excluding those still living in Java or Bali five years earlier and those born after settlement, by the settlers the program placed

[< back](#)

Backup: what the migration evidence does and does not say

	Birthplace decomposition (% of 2010 cohort)			
	(1) Same district	(2) Same province	(3) Other outer-island province	(4) Java or Bali
TMV AgSim (z, 300min)	-0.043* (0.023)	0.043** (0.018)	0.018 (0.021)	-0.018* (0.011)
Transmigrant count (IDW)	✓	✓	✓	✓
Baseline Pop 1980	✓	✓	✓	✓
Urban share 1980	✓	✓	✓	✓
Total Empl 1980	✓	✓	✓	✓
Agri Empl 1980	✓	✓	✓	✓
Agro-Clim. TMV controls	✓	✓	✓	✓
Mean dep var	0.667	0.130	0.116	0.087
Std dev dep var	0.187	0.092	0.129	0.084
Observations	133	133	133	133

- ▶ The four birthplace shares sum to one over each city's 1955-1990 birth cohort, so the coefficients sum to zero
- ▶ The same-district share falls 4.3 percentage points and the same-province share rises 4.3 percentage points
- ▶ The Java or Bali share falls 1.8 percentage points; the other-outer-island share does not move detectably

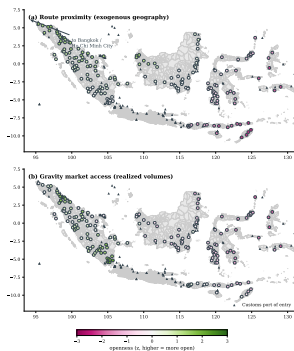
Backup: migration in the model, and urban emergence

- ▶ Migration enters the model as an estimated friction, disciplined by two pieces of evidence
 - ▷ A gravity estimate from the IFLS panel pins the composite moving cost
 - ▷ The IFLS never-mover share is 0.533, reported as an untargeted over-identifying moment because coverage limits make its level a lower bound on true mobility
- ▶ Urban emergence: exposure thickens existing cities; the extensive response is small and marginal

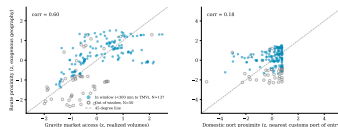
◀ back

Backup: how trade openness is measured

Route geography, predetermined



Route against market access and port proximity



- ▶ Route distance: nearest customs port plus the sea leg to Bangkok or Ho Chi Minh City (Bazzi, 2017)
- ▶ Rutz (1987) places the anchor port of 124 of the 133 cities in the 1986 harbor inventory or earlier
- ▶ Validated against the price response to the 2004 rice-import ban

The same split on baseline route distance to the rice hubs

	(1) ln pop	(2) ln emp	(3) unemp rate	(4) ln L agric	(5) agric %	(6) ln L modern	(7) modern %
AgSim, more closed cities: β_1	0.130 (0.086)	0.113 (0.082)	0.003 (0.009)	-0.152 (0.132)	-0.065*** (0.024)	0.283*** (0.090)	0.093*** (0.022)
AgSim, more open cities: $\beta_1 + \beta_2$	0.020 (0.057)	-0.011 (0.059)	0.012* (0.007)	-0.177* (0.096)	-0.046** (0.021)	0.107 (0.067)	0.054*** (0.016)
High openness (=1)	0.023 (0.067)	0.005 (0.082)	-0.005 (0.011)	0.044 (0.150)	0.003 (0.028)	-0.012 (0.084)	-0.013 (0.019)
Mean dep var	11.839	10.930	0.123	9.816	0.374	10.229	0.517
Std dev dep var	0.858	0.843	0.041	0.898	0.184	0.979	0.157
Observations	133	133	133	133	133	133	133

Note: High openness = below-median route distance to the Thai and Vietnamese rice hubs. Main Spec throughout: settler count, 1980 city baselines and the agro-climatic controls. Conley 200 km

- Route distance is predetermined trade geography: the port network plus the sea leg to Bangkok or Ho Chi Minh City
- Identification relies on route distance; market access, in part an equilibrium outcome, is additional evidence
- Closed cities respond 1.4 times as much as open ones on the agricultural share, 2.6 times on modern employment
- The closed-open gap is significant at 10% on modern employment, imprecise on the agricultural share

Backup: it is not public provision either

	INPRES primary schooling				INPRES public latrines	Placebo: the schools INPRES did not build
	Share of schools (1)	Schools per 1,000 people (2)	Share of teachers (3)	Share of buildings (4)	Share of latrines (5)	Schools per 1,000 people (6)
Agro-climatic exposure (z)	0.0307 (0.0275)	0.0093 (0.0286)	0.0098 (0.0307)	0.0380 (0.0265)	0.0343 (0.0737)	-0.1328* (0.0737)
<i>p-value</i>	0.264	0.745	0.750	0.152	0.642	0.072
Mean of outcome	0.3360	0.3927	0.3249	0.3385	0.4145	0.7960
Observations	119	119	119	119	108	119

- ▶ The alternative reading is that the state, having settled the villages, then built the roads and schools that transformed the cities
- ▶ Exposure does not predict 1990-2007 road upgrades, the facilities a city acquired, or where INPRES built schools
- ▶ Public provision also gives a city's rice-import connections no role, so it cannot produce the openness split

Backup: the model, element by element

- ▶ **Food price.** Each location buys food from every other at an iceberg cost rising in cost-distance, and the delivered price is a CES aggregate over sources, so a productive hinterland lowers the price its city pays
- ▶ **Modern good.** Labor and housing land; the headline calibration sets its agglomeration term to zero
- ▶ **Land.** Farming and housing share a fixed endowment, so agriculture can shrink through land as well as labor
- ▶ **Sector choice.** θ_S governs how sharply workers sort between agriculture and the modern sector within a location
- ▶ **Location choice.** θ_L governs sorting across locations on real income net of amenities
- ▶ **Closing.** Labor and land markets clear; under the closed rebate no city claims another location's output

The channel decomposition runs on a variant that splits the modern composite into a catchment-traded and a purely local component. With one tradable good and a closed land-rent rebate, no city can finance a persistent net food purchase

Backup: migration enters as an estimated friction

- ▶ At the estimated θ , moving costs borne from 1980 origins **hold** people near the host villages, so the population response comes out wrong-signed; the channel diagnosis is unchanged
- ▶ The root cause is a horizon mismatch. The gravity kernel prices a move per decade; the data ask about cumulated mobility over thirty years
- ▶ So migration is calibrated as an estimated friction, consistent with the high internal moving costs in [Bryan and Morten \(2019\)](#), and a fully dynamic per-period model is the natural extension

◀ back

Backup: why Stone-Geary preferences

- ▶ The agricultural contraction runs through the subsistence floor. Cheaper food frees supernumerary income, and the share spent on food falls
- ▶ A PIGL variant was explored as a comparison and is not part of the paper
- ▶ The floor is also the only non-homotheticity the model keeps, so the demand side stays legible: one parameter, one economic reading

◀ back

Backup: the estimator, step by step

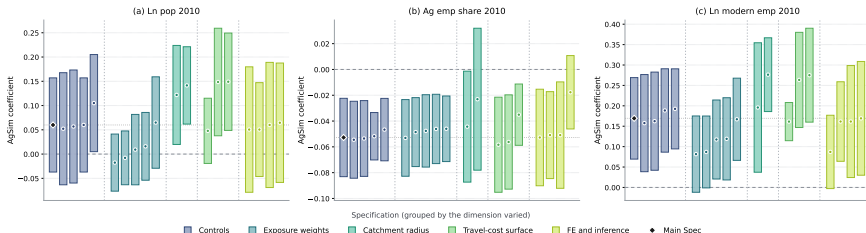
1. Run the Main Spec regression on the **data**, giving the auxiliary coefficients that are the estimation targets
2. At a candidate (θ_L, θ_S) , solve the model, apply the productivity shock, and generate a model panel of the same cities
3. Run the **same regression, with the same controls**, on the model panel
4. Both sides are put in z - z units so the model and data slopes are directly comparable
5. Minimize the standard-error-weighted distance between the two coefficient vectors

Minimum-distance GMM on auxiliary-regression coefficients (indirect inference): data and model target the same object

Two parameters, two targeted moments: exactly identified, so fit is judged on the untargeted moments

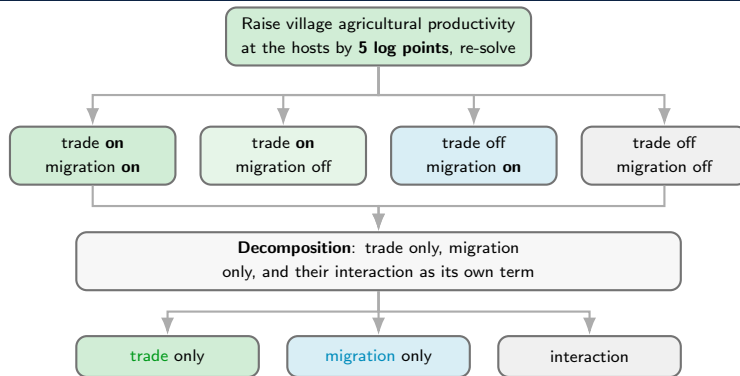
Backup: no specification reverses the compositional result

Specification curve across 19 specifications, three headline outcomes



- ▶ Across 19 specifications, none is both wrong-signed and distinguishable from zero for any headline outcome
- ▶ The agricultural share clears the 10% level in 17 of 19, modern employment in 16, log population in 5
- ▶ Borusyak and Hull, controlling for expected exposure: agricultural share -0.042 at 5%, modern share $+0.051$ at 1%
- ▶ Model side: the decomposition uses traded services; the single-tradable baseline cannot reach the two targets

Counterfactuals: shut trade or migration, one at a time



- ▶ Trade switches off by pushing trade cost toward autarky, migration by fixing each location's population
- ▶ The two contributions need not sum to the total because the channels interact, so all four states are solved
- ▶ The interaction is reported, never assigned to either channel
- ▶ These shares describe the calibrated model, never the reduced-form estimates

Identification threats and their answers

Threat	The answer in the design	Where
Sites were placed near promising cities	Placement and scale are held fixed: settler counts and village agro-climate are controls; the identifying variation is the origin-destination match	Main Spec
AgSim proxies good destinations, not the match	Destination agro-climatic conditions enter directly; villages on identical land differ only in who landed there	Main Spec
The settlers transform the cities themselves	Farms to farms by design; the settlers stay; the Java- and Bali-born share in cities falls	Result 2
Exposed cities were already on different paths	Exposure is uncorrelated with most 1980 population, employment, and composition levels; the two baselines it flags are controls	Placebo
Exposed cities gain at their neighbors' expense	The model aggregates city and hinterland jointly, so the counterfactual reads the region; in-migrants were born in the same province	Result 2, CF
133 spatial units	Conley 200 km; no specification in the curve is both wrong-signed and significant at 10%	Robustness